Course Name: Blockchain and its Applications (NOC26_CS34)

Assignment 11 - Week 11 (March 2026)

TYPE OF QUESTION: MCQ/MSQ

Number of questions: **10** **Total mark: 10 X 1 = 10**

QUESTION 1

What specific gap does Hyperledger Aries fill when used alongside Hyperledger Indy?

a. It provides the consensus algorithm necessary for Indy to mine blocks.
b. It introduces a new native cryptocurrency to incentivize Indy node operators.
c. It gives the infrastructure for blockchain-routed peer-to-peer interactions and the secure transport of digital credentials between participants.
d. It replaces Indy's distributed ledger entirely by acting as a faster, centralized database.

Answer: (c)

Detailed solution:
Aries solves the transport gap by providing peer-to-peer interaction infrastructure for credentials.
Please refer to week 11 lecture 51.

QUESTION 2

According to the architecture of the Aries Cloud Agent Python (ACA-Py) discussed in the lecture, which of the following are primary modules/components?

a. Core Capabilities
b. REST API Module
c. Proof of Work Consensus Engine
d. Aries Protocols (DIDcomm)

Answer: (a), (b) and (d)

Detailed solution:
Please refer to week 11 lecture 51.

QUESTION 3

In the context of selfish mining, suppose an attacker's mining pool privately mines two consecutive blocks while the honest network mines one competing block. The attacker then releases its private chain, and both attacker blocks are successfully accepted into the main chain (i.e., no fork race or block loss occurs). If the block reward is 6.25 bitcoins per block, how much total reward does the attacker's pool earn?

a. 6.25 bitcoins
b. 12.5 bitcoins
c. 18.75 bitcoins
d. 8.25 bitcoins

Answer: (b)

Detailed solution:
the attacker's pool successfully replaces the public chain with its own longer chain containing two blocks, which means both of its blocks are accepted while the honest miner's block is discarded. Since each block gives a reward of 6.25 bitcoins, the attacker earns a total of $2 \times 6.25 = 12.5$ bitcoins.
Please refer to week 11 lecture 53.

QUESTION 4

In the context of transaction privacy leakage, what mechanism is mentioned in the lecture as a way to use decoys to obscure the trace of cryptocurrency transactions?

a. Genesis Blocks
b. Chaff coins (Mixins)
c. 51% Padding
d. Webhooks

Answer: (b)

Detailed solution:
Chaff coins/Mixins are used as decoys to prevent transaction privacy leakage.

Please refer to week 11 lecture 52.

QUESTION 5

In an eclipse attack scenario discussed in lecture 54, suppose an attacker controls 40% of the network's hashing power, while a victim node has 30% and the rest of the network holds the remaining 30%. If the attacker successfully isolates (eclipses) the victim from the rest of the network, which of the following statements correctly describe what happens?

a. The attacker now effectively controls 70% of the hashing power on the main chain.
b. The honest nodes (victim + rest of the network) can no longer combine their 60% hashing power.
c. The attacker directly competes with the victim with a 40% vs 30% advantage.
d. The attacker directly competes with the rest of the network with a 40% vs 30% advantage.

Answer: (b), (c), and (d)

Detailed solution:
When the attacker eclipses the victim, the network is split into separate parts that cannot communicate with each other. Even though the honest nodes together have 60% hashing power, they can't coordinate or build on each other's blocks anymore. As a result, the attacker effectively competes against each group separately, having a 40% vs. 30% advantage over both the victim and the rest of the network.

QUESTION 6

In the Fomo3D smart contract game, every time someone buys a ticket, the countdown timer is extended by 30 seconds. Suppose an attacker buys a ticket when there are exactly 3 minutes left on the clock, and then launches a suppression attack by flooding the network with high gas-price transactions so that no one else's transactions get included in blocks. Assuming no honest transactions go through during this time, how many ticket purchases did the attacker effectively prevent?

a. 3 purchases
b. 6 purchases
c. 10 purchases
d. 30 purchases

Answer: (b)

Detailed solution:
Since 3 minutes equals 180 seconds and each ticket would extend the timer by 30 seconds, up to 6 purchases could have occurred in that time (180 ÷ 30 = 6). By blocking all other transactions, the attacker prevents these 6 possible extensions, allowing the timer to run out and ensuring their win. Please also refer to week 11 lecture 54.

QUESTION 7

What is the primary objective of an "Eclipse Attack" on a blockchain network?

a. To gain majority control (51%) of the network's hashing power by combining mining pools
b. To isolate a target node from the rest of the network by filling its peer connections with attacker-controlled nodes
c. To take advantage of smart contract bugs to steal funds
d. To crack a user's private key using brute force

Answer: (b)

Detailed solution:
An Eclipse Attack is all about isolating a node. The attacker takes control of all the incoming and outgoing connections of the victim node, so it only communicates with malicious peers instead of honest ones. This isolation allows the attacker to control what information the victim sees, which can then be used to carry out further attacks like double-spending or selfish mining.

QUESTION 8

In an eclipse attack, an attacker can overwhelm a node by filling its "new" and "tried" peer tables with malicious IP addresses, effectively isolating it from honest peers. Based on the lecture, which of the following is a practical way developers can reduce this risk?

a. Force all nodes to rely on a centralized and trusted DNS server for peer discovery.
b. Change how IP address buckets are managed so that malicious entries are gradually replaced by honest ones.
c. Require nodes to pass a strict Proof of Stake check before joining the network.
d. Encrypt the node's local database to hide the peer tables from attackers.

Answer: (b)

Detailed solution:
A good defense is to redesign how peer tables are updated and maintained, ensuring that attacker-controlled IP addresses don't permanently dominate the tables. By allowing honest IPs to eventually replace malicious ones, the node can recover from isolation attempts and reconnect with the legitimate network.

QUESTION 9

In a double-spending attack where the attacker is working together with a mining pool, how is the vendor actually cheated?

a. The attacker sends a payment to the vendor and receives the service, but the colluding mining pool later confirms a conflicting transaction that sends the same coins back to the attacker.
b. The attacker steals the vendor's digital signature and uses it to drain their wallet.
c. The attacker creates a private chain of blocks that removes the vendor's node from the network.
d. The attacker uses fake "chaff coins" to make the payment appear larger than it really is.

Answer: (a)

Detailed solution:
In this type of attack, the attacker first sends a legitimate-looking transaction to the vendor and gets the goods or service. Meanwhile, they secretly create another transaction that sends the same coins back to themselves. Because they are collaborating with a mining pool, that pool gives priority to the attacker's conflicting transaction and includes it in the blockchain. As a result, the transaction paying the vendor is invalidated, meaning the vendor ends up with nothing while the attacker keeps both the goods and the money.

QUESTION 10

In an Eclipse Attack, how does the attacker actually isolate a victim node from the rest of the network?

a. By interfering with the ISP's routing hardware
b. By launching a DDoS attack on the victim's internet provider
c. By hacking the victim's wallet so it ignores blocks
d. By filling the victim's peer tables with attacker-controlled IPs

Answer: (d)

Detailed solution:
The attacker takes advantage of how the node stores peer IP addresses. They keep adding their own controlled IPs into the victim's "new" and "tried" tables. Over time, these tables get filled with attacker nodes, so when the victim connects to peers, it ends up connecting only to the attacker's machines. This cuts it off from the honest network without needing anything like ISP attacks or software hacks.